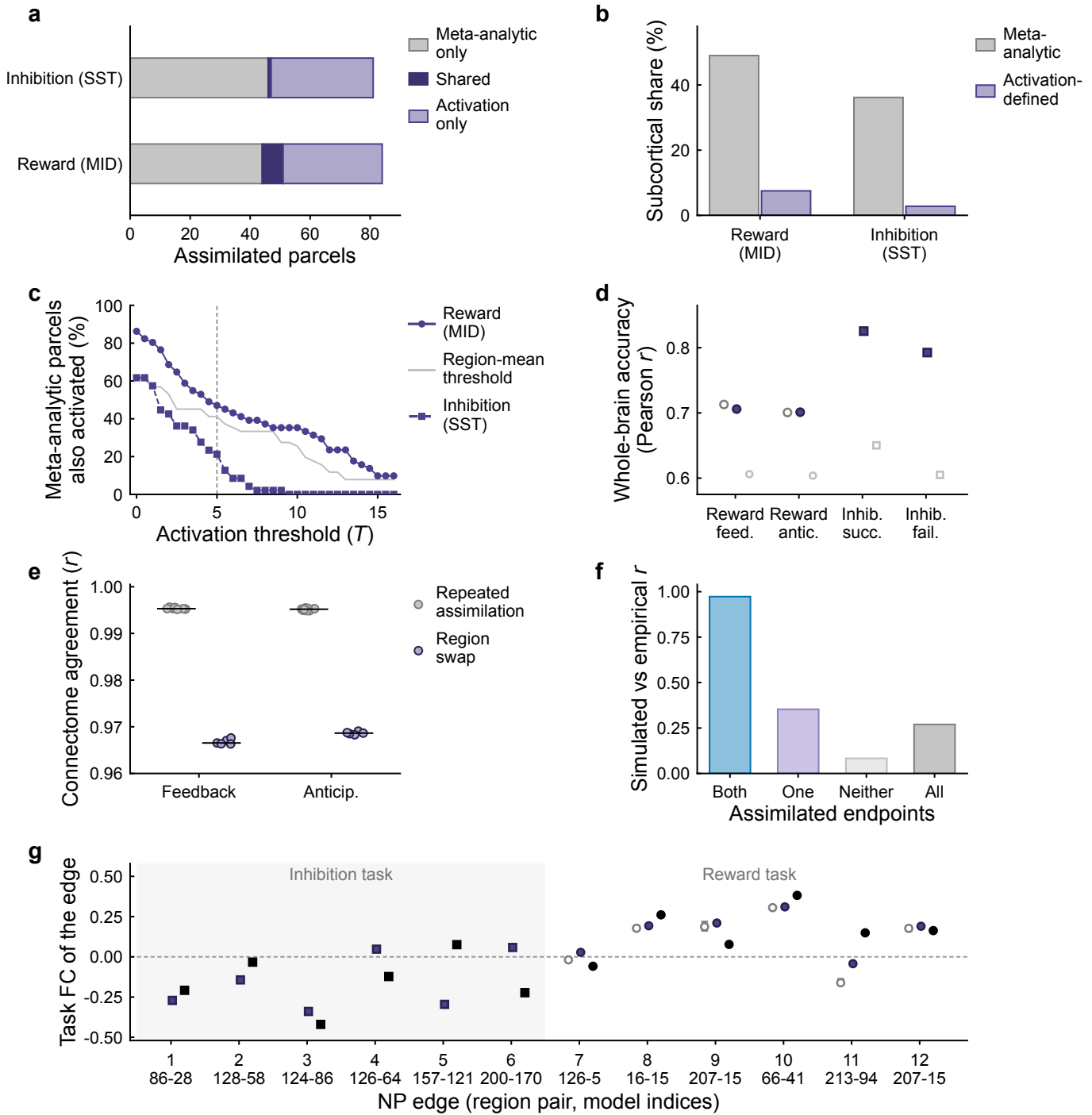




Supplementary Fig. S22 | Meta-analytic versus activation-defined assimilation regions. One activation-defined region set was built per task from the cohort's group-level task T maps and matched to the meta-analytic set on assimilated voxel count (reward set, 2,013 versus 2,022 voxels; inhibition set, 2,185 versus 1,959). Both were then assimilated and simulated in one participant's 100-million-neuron voxel-wise digital twin brain and scored against that participant's empirical task functional connectivity over all 23,436 edges of the 217-region model. The meta-analytic set was assimilated five independent times on the reward task, which provides the run-to-run envelope against which the region swap is judged; no 100-million-neuron meta-analytic assimilation of the inhibition task exists for this participant, so its only comparator is the 10-million-neuron cohort model, which differs in scale as well as in region definition and is shown as context rather than as a matched comparison. One participant with repeated assimilations is a set of repeated measurements rather than a sample, so every panel is descriptive: no error indicator other than the observed run-to-run range, and no inferential test. Panels carry no numbers; every value is given below, and the marker key under **g** applies to **d** and **g**. **a**, Parcel composition of the two simulated sets in the participant's own voxel space, as meta-analytic only / shared / activation only: 44 / 7 / 33 parcels on the reward task and 46 / 1 / 34 on the inhibition task, so the sets share 7 of 51 meta-analytic parcels on the reward task (Dice 15.4%, 17.4% by voxel) and 1 of 47 on the inhibition task (2.4%, 1.5% by voxel). **b**, Subcortical share of each set in the 217-region model space (reward, 25 of 51 regions subcortical for the meta-analytic set versus 3 of 40 for the activation-defined set; inhibition, 17 of 47 versus 1 of 36); the activation-defined sets load onto visual and motor cortex instead. **c**, Percentage of meta-analytic parcels that also show positive group-level activation, as the activation threshold is raised (solid line and circles, reward; dashed, inhibition); the pale lines apply the threshold to the region mean rather than to any voxel). The definitions agree at liberal thresholds (86% of reward and 62% of inhibition parcels at $T > 0$) and diverge as the threshold rises (47% and 21% at $T \geq 5$). **d**, Whole-brain accuracy for the four task conditions. Replacing the meta-analytic regions that did not improve it: reward feedback-hit $r = 0.7059$ versus 0.7129 ± 0.0007 (range 0.7120-0.7138) over the five meta-analytic runs, a change of -0.0070, about ten between-run standard deviations but -1.0% of r ; reward anticipation-hit 0.7011 versus 0.7007 ± 0.0006 (+0.0004, +0.1%). The inhibition conditions reach 0.8255 and 0.7926 with the activation-defined set at 100 million neurons against 0.6502 and 0.6050 for the 10-million-neuron meta-analytic cohort model, a difference that cannot be attributed to the region definition because the two runs differ in scale. **e**, Agreement between pairs of simulated connectomes, one point per pair and the bar at the median: the five repeats of the meta-analytic set reproduce each other at $r = 0.9949$ - 0.9956 (10 pairs per condition) whereas the activation-defined run reproduces them at $r = 0.9663$ - 0.9691 (5 pairs per condition), so the region swap perturbs the connectome about 6.8 times more than rerunning the same assimilation - and accuracy still did not improve. **f**, Simulated-empirical agreement for all edges of the meta-analytic 100-million neuron baseline, grouped by how many of the edge's two endpoints were assimilated ($r = 0.9729$, $n = 1275$, $MSE = 0.00671$; $r = 0.3532$, $n = 8466$, $MSE = 0.07995$; $r = 0.0832$, $n = 13695$, $MSE = 0.09875$; $r = 0.2698$, $n = 23436$, $MSE = 0.08695$). **g**, The twelve NP edges. Only edges 7 and 11 fall outside the five-run meta-analytic range ($z = 6.1$ and 8.3 in units of the between-run standard deviation), and both are edges whose endpoints the meta-analytic set assimilated and the activation-defined set dropped - the same pattern as **f**. Edges 1-6 belong to the inhibition task and have no meta-analytic 100-million-neuron comparator.